

Bank of Leeds Strategic Infrastructure White Paper (2031)

Yorkshire Integrated Rail Network (YIRN)

Building Europe's Most Connected Regional Railway

Prepared for:

Bank of Leeds

Department of Strategic Infrastructure & Economic Development

Programme Value

£28 Billion

Delivery Period

2032–2050

Funding Model

Public–Private Partnership (PPP/PFI-style concession with government participation)

Executive Summary

By 2031, Yorkshire has become one of Britain's fastest-growing regional economies, supported by advanced manufacturing, finance, energy, tourism and technology. However, rail infrastructure—much of it originating in the Victorian era—has become the principal constraint on further growth.

The Bank of Leeds proposes the creation of the **Yorkshire Integrated Rail Network (YIRN)**: a unified programme of new lines, upgraded corridors, electrification, freight improvements and integrated ticketing.

Unlike previous rail programmes focused on isolated schemes, the YIRN is conceived as a **regional economic system**. Railways are treated as productive infrastructure that expands labour markets, supports housing development, reduces road congestion and raises land values. Existing regional proposals have similarly argued that stronger rail connectivity can unlock substantial economic growth, employment and housing across Yorkshire.

The strategic objective is:

"No major Yorkshire town should be more than one hour from Leeds by rail."

The Anglo-Futurist Railway Philosophy

The railway is regarded as the backbone of Yorkshire's economy.

Investment is based on five principles:

- **Speed** – faster inter-city travel.
 - **Frequency** – turn-up-and-go services.
 - **Reliability** – modern signalling and electrification.
 - **Beauty** – stations as civic landmarks.
 - **Productivity** – every rail investment must stimulate commercial development.
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Programme Value

Programme	Investment
New railway construction	£9.5bn
Existing line upgrades	£5.5bn
Electrification	£3.5bn
Leeds Rail Hub	£2.5bn
Stations	£2.0bn
Freight network	£2.0bn
Digital signalling	£1.5bn
Rolling stock	£1.5bn

Total Investment

£28 Billion

Network Structure

Rather than a collection of disconnected routes, Yorkshire is organised into four strategic corridors.

White Rose Express

High-speed regional spine

Hull

↓

York

↓

Leeds

↓

Wakefield

↓

Sheffield

Target speed:

160–200 km/h

Pennine Express

Leeds

↓

Bradford

↓

Huddersfield

↓

Manchester

Designed to complement wider Northern Powerhouse Rail improvements and upgraded TransPennine infrastructure already proposed by government.

Target journey:

Leeds–Manchester

25–30 minutes

Yorkshire Coast Line

Leeds

↓

York

↓

Scarborough

↓

Bridlington

↓

Hull

Fast regional coastal services.

Yorkshire Dales Line

Leeds

↓

Skipton

↓

Bolton Abbey

↓

Settle

↓

Carlisle

Expanded tourism and freight.

Leeds Grand Central

Investment

£2.5 Billion

Leeds becomes the principal railway interchange in northern England.

Facilities include:

- underground through platforms,
 - high-capacity concourse,
 - airport express platforms,
 - retail district,
 - integrated bus and tram interchange.
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Journey Time Targets

Route	Today	2050 Target
Leeds–Sheffield	~40 min	22 min
Leeds–York	~25 min	15 min
Leeds–Hull	~55 min	35 min
Leeds–Scarborough	~80 min	50 min

Route	Today	2050 Target
Leeds–Bradford	~20 min	10 min
Leeds–Harrogate	~35 min	18 min
Leeds–Manchester	~50–55 min	25–30 min
Sheffield–Hull	~90 min	50 min

These targets broadly align with the long-term direction of Northern Powerhouse Rail and the wider Integrated Rail Plan, which emphasise electrification, digital signalling, additional tracks and faster regional services.

Yorkshire Metro Services

Major urban areas receive high-frequency suburban rail.

Frequency:

Every 10 minutes.

Cities include:

- Leeds
- Sheffield
- Bradford
- York
- Hull

Integrated with local tram and bus systems.

Freight Renaissance

Investment

£2 Billion

The Bank of Leeds considers freight one of the largest economic opportunities.

Projects include:

- electrified freight routes,
- upgraded intermodal terminals,
- improved links to Humber ports,
- logistics parks,
- overnight freight corridors.

Target:

Increase rail freight market share from road, reducing motorway congestion while supporting manufacturing exports.

Property Investment Strategy

The Bank establishes the:

Yorkshire Rail Property Fund

Initial Capital

£3 Billion

Acquire land around:

- Leeds
- Bradford
- Sheffield
- York
- Hull
- Harrogate
- Wakefield

Develop:

- offices,
- apartments,
- hotels,
- science parks,
- retail.

Transport-oriented development allows part of the increase in land values created by new stations to help finance infrastructure.

Financing Model

Yorkshire Infrastructure Partnership

The proposal adopts a concession-based public-private partnership rather than relying entirely on public capital.

Government

- statutory approvals,
- land assembly,
- strategic grants.

Bank of Leeds

- long-term infrastructure finance,
- bond issuance,
- property investment.

Pension Funds

- long-duration debt.

Private Sector

- station development,
- commercial property,
- logistics facilities.

This approach is intended to allocate construction, financing and operational risks contractually, recognising that PPP/PFI models require careful governance and transparent risk allocation to avoid poor value for money.

Yorkshire Rail Bond

Bond Issue

£20 Billion

Term

40 years

Expected Investors:

- UK pension funds,
- insurance companies,
- sovereign wealth funds,
- infrastructure funds.

Repayment sources include:

- access charges,
 - station retail,
 - commercial property,
 - land value capture,
 - government availability payments.
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Economic Impact

Construction

Approximately

220,000 job-years

Permanent Employment

Rail operations

12,000

Commercial development

25,000

Hospitality

15,000

Retail

18,000

Professional services

10,000

Total

Approximately

80,000 permanent jobs

Economic Benefits

Independent regional analyses have estimated that improved rail connectivity across Yorkshire could support around **£20 billion of additional gross value added over a decade**, alongside significant housing and employment growth.

Within the Bank of Leeds scenario, the mature network is expected to contribute through:

- larger labour markets,
- faster business travel,
- higher tourism,
- lower logistics costs,
- transit-oriented development.

Illustrative long-term regional impact:

Category	Annual Value
Productivity gains	£5.0bn
Tourism	£1.5bn
Property development	£2.0bn
Freight savings	£1.0bn
Business investment	£2.5bn
Estimated annual economic uplift:	

Approximately £12 Billion

Delivery Programme

Phase I (2032–2037)

- electrification,
- digital signalling,
- station modernisation,
- Leeds Hub.

Investment:

£8 billion

Phase II (2037–2043)

- new inter-city corridors,
- freight upgrades,
- airport connections.

Investment:

£10 billion

Phase III (2043–2050)

- remaining new lines,
- capacity expansion,
- property developments.

Investment:

£10 billion

Final Investment Thesis

The Bank of Leeds views rail not simply as a transport service, but as **economic infrastructure**.

Every minute removed from a journey expands labour markets. Every new station creates opportunities for housing and commercial investment. Every freight train moved from motorway to railway lowers logistics costs and improves productivity.

The Yorkshire Integrated Rail Network therefore aims to create a region where movement between cities is fast, frequent and reliable, allowing Yorkshire to function as a single integrated economy rather than a collection of separate urban centres.

The long-term vision is clear:

One Region. One Network. One Yorkshire Economy.